

ECON 712 - Answer Key

Q1: Linear Transformation of variables

To answer this question, the key is to realize the following:

$$\begin{aligned} M_L X &= (I - L(L^T L)^{-1} L^T) X \\ &= X - \frac{1}{n} \begin{bmatrix} 1 & \dots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} x_{11} & \dots & x_{1k} \\ x_{21} & \dots & \vdots \\ \vdots & \ddots & \vdots \\ x_{n1} & \dots & x_{nk} \end{bmatrix} \\ &= X - L \bar{X}^T = W \end{aligned}$$

Similarly,

$$\begin{aligned} M_L y &= (I - L(L^T L)^{-1} L^T) y \\ &= y - \frac{1}{n} \begin{bmatrix} 1 & \dots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \\ &= y - L \bar{y} = Z \end{aligned}$$

From the regression model:

$$M_L y = M_L L_n \hat{\gamma} + M_L X \hat{\beta} + M_L M_{LX} y$$

So,

$$Z = W \hat{\beta} + M_{LX} y$$

We proceed as we did in class to prove the FWL theorem,

$$X^T M_L Z = X^T M_L W \hat{\beta} + X^T M_L M_{LX} y$$

$$(M_L X)^T Z = (M_L X)^T W \hat{\beta} + (M_{LX} X)^T y$$

$$W^T Z = W^T W \hat{\beta}$$

$$W (W^T W)^{-1} W^T Z = W (W^T W)^{-1} W^T W \hat{\beta}$$

$$P_W Z = W \hat{\beta}$$

Therefore, $\hat{\beta} = \hat{\alpha}$

To show that residuals are equal, we use

$$Z - W \hat{\beta} = M_{LX} y$$

since $\hat{\beta} = \hat{\alpha} \Rightarrow Z - W \hat{\alpha} = M_{LX} y$

$$M_W Z = M_{LX} y$$

QII: Finite Sample properties of OLS

$$\hat{\beta} = (Z^T Z)^{-1} Z^T S$$

substituting for Z and S

$$\hat{\beta} = \frac{b}{a} (X^T X)^{-1} X^T y$$

$$\hat{\beta} = \frac{b}{a} (X^T X)^{-1} X^T (X\beta + u)$$

$$\hat{\beta} = \frac{b}{a} \beta + \frac{b}{a} (X^T X)^{-1} X^T u$$

$$E[\hat{\beta} | X] = \frac{b}{a} \beta + \frac{b}{a} (X^T X)^{-1} X^T E[u | X]$$

$$E[\hat{\beta} | X] = \frac{b}{a} \beta$$

$$\text{by LIE} \quad E[\hat{\beta}] = \frac{b}{a} \beta = \beta + \underbrace{\left(\frac{b}{a} - 1\right)}_{\text{Bias}} \beta$$

Note that if $b = a \Rightarrow \text{Bias} = 0$

QIII: Linear Regression and selection bias

(a)

Let $X_i = [1 \ D_i]^T$

$$X_i y_i = \begin{bmatrix} y_i \\ D_i y_i \end{bmatrix} \Rightarrow E[X_i y_i] = \begin{bmatrix} \sum_{i=1}^N y_i \\ \sum_{i=1}^N D_i y_i \end{bmatrix}$$

$$X_i X_i^T = \begin{bmatrix} 1 & D_i \\ D_i & D_i^2 \end{bmatrix} \Rightarrow E[X_i X_i^T] = \begin{bmatrix} N & N_1 \\ N_1 & N_1 \end{bmatrix}$$

where $N = N_0 + N_1$

$$\Rightarrow E[X_i X_i^T]^{-1} = \frac{1}{NN_1 - N_1^2} \begin{pmatrix} N_1 & -N_1 \\ -N_1 & N \end{pmatrix}$$

$$\beta^{OLS} = \frac{1}{NN_1 - N_1^2} \begin{bmatrix} N_1 & -N_1 \\ -N_1 & N \end{bmatrix} \begin{bmatrix} \sum_{i=1}^N y_i \\ \sum_{i=1}^N D_i y_i \end{bmatrix}$$

$$\beta^{OLS} = \begin{bmatrix} \frac{[N_1 \sum y_i - N_1 \sum D_i y_i]}{NN_1 - N_1^2} \\ \frac{[-N_1 \sum y_i + N \sum D_i y_i]}{NN_1 - N_1^2} \end{bmatrix}$$

$$\beta^{OLS} = \begin{bmatrix} \frac{\sum (y_i - D_i y_i)}{N_0} \\ \frac{\sum y_i}{N_1} - \frac{[\sum (y_i - D_i y_i)]}{N_0} \end{bmatrix} = \begin{bmatrix} E[y_i | D_i = 0] \\ \underbrace{E[y_i | D_i = 1] - E[y_i | D_i = 0]}_{\beta_1} \end{bmatrix}$$

Which is equivalent to comparing conditional expectations across bins

(b)

$$\beta_1 = E[y_{1i} | D_i = 1] - E[y_{0i} | D_i = 0]$$

$$\beta_1 = \underbrace{E[y_{1i} - y_{0i} | D_i = 1]}_{\text{ATT} = \text{ATE} = \rho} + \left[E[y_{0i} | D_i = 1] - \right.$$

ATT = ATE = ρ

Because the
effect is
constant

$E[y_{0i} | D_i = 0]$

Selection
Bias

c)

Selection Bias

$$\beta_1 = \underbrace{E[P_i]}_{\text{ATE}} + \underbrace{\left(E[y_{0i} | D_i=1] - E[y_{0i} | D_i=0] \right)}_{\text{Additional effect on treated individuals}} + \underbrace{\left(E[P_i | D_i=1] - E[P_i] \right)}$$

Additional effect on
treated individuals

Q IV: Goodness of Fit

$$P_X Z = P_X (y - cL) = P_X y - c P_X L = P_X y - cL$$

$$\text{So } R_u^2 = \frac{\|P_X y - cL\|^2}{\|y - cL\|^2}$$

On the other hand,

$$M_L Z = M_L (y - cL) = M_L y - c M_L L = M_L y$$

So R_c^2 remains the same

QV: Conditional Expectation Function

(a)

$$E[y|X=0] = \text{Prob}(y=1|X=0) = \frac{0.6}{0.8} = \frac{3}{4}$$

$$E[y|X=1] = \text{Prob}(y=1|X=1) = \frac{0.1}{0.2} = \frac{1}{2}$$

(b)

$$m(x) = \frac{3}{4} - \frac{1}{4}x$$

$$(c) \quad u = y - \left(\frac{3}{4} - \frac{1}{4}x\right)$$

$$E[u|X=0] = \left(1 - \frac{3}{4}\right) \frac{3}{4} + \left(0 - \frac{3}{4}\right) \left(1 - \frac{3}{4}\right) = 0$$

$$E[u|X=1] = \left(1 - \frac{1}{2}\right) \frac{1}{2} + \left(0 - \frac{1}{2}\right) \frac{1}{2} = 0$$

By LIE

$$E[u] = E[E[u|x]] = E[0] = 0$$

Since $E[u] = E[u|x] = 0$ for $x = 0, 1$,

u is mean independent from x . Therefore,

$$\text{cov}(x, u) = 0$$